

Detecting Interictal Epileptiform Discharges in EEG using Functional Data Analysis

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Research Project (PSGY 4012)

Under the Supervision of: Mark van Rossum

Word Count: 7963

Abstract

Interictal epileptiform discharges (IEDs) are brief transients seen in EEG that support the diagnosis of epilepsy, but their variable morphology and similarity to non-epileptic transients make automated detection difficult. Conventional approaches typically use scalar features, discarding information about the shape of the waveform. This study investigates whether Functional Data Analysis (FDA) can provide a useful representation of IED morphology and support automated detection in continuous scalp EEG. We developed a four-layer detection system comprising candidate generation, event consolidation, FDA-based classification, and a final decision layer that uses spatial information. We represented candidate waveforms as time-domain functions using B-spline basis expansion. We then applied Fisher–Rao registration and functional principal component analysis (FPCA), using the resulting functional principal components in logistic regression for classification. Within the complete detection pipeline, the final layer reached a sensitivity of 0.72 at an operating point of no more than 10 false positives per minute, with an AUROC of 0.87 and a PR-AUC approximately 20 times the chance-level baseline. In external testing, detection rate did not reliably separate confirmatory from non-confirmatory recordings (AUROC = 0.58), whereas the mean score of the ten highest-scoring events provided greater separation (AUROC = 0.72). However, external recordings had labels only at the recording level, preventing assessment of individual detections. These results indicate that functional representation can provide discriminative information for IED detection, while further development is needed to improve generalisation and support reliable automated detection in continuous EEG.

Keywords: Functional Data Analysis, IED Detection, Scalp EEG, Functional Principal Component Analysis, B-Splines

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Introduction

Epilepsy is one of the most common neurological conditions, affecting approximately 51.7 million people globally (Feigin et al., 2025). The International League Against Epilepsy (ILAE) defines epilepsy as an enduring predisposition to generate seizures (Fisher et al., 2014). A seizure arises from an imbalance between excitatory and inhibitory drive in the brain, producing hypersynchronous neuronal discharge in which spatially distributed neuronal populations fire action potentials simultaneously (McCormick & Contreras, 2001).

Clinicians call the interval between epileptic seizures the interictal period. During this interval, the EEG often shows brief sharp transients called interictal epileptiform discharges (IEDs). IEDs result from an abnormal cellular phenomenon called paroxysmal depolarisation shift (PDS). A PDS is a large, sustained shift in membrane potential, lasting from 40 to over 400 ms and shifting the voltage by 30 mV or more, with a burst of high-frequency action potentials riding on its peak. The term paroxysmal describes this pattern of sudden onset and short duration. When a local network of thousands of adjacent neurons fires these PDS events in tight synchrony, their combined intracellular voltage fluctuations produce the distinct, sharp voltage deflection seen as an IED on the scalp or cortical surface EEG. (de Curtis & Avanzini, 2001; Matsumoto & Ajmone Marsan, 1964). IEDs are classified by duration as spikes (20-70 milliseconds) or sharp waves (70-200 milliseconds), and are frequently followed by a slow after-wave (demonstrative IED shown in Figure 1).

These discharges transiently disrupt memory and attention even without a clinical seizure (Aarts et al., 1984; Holmes & Lenck-Santini, 2006). The ILAE criterion described earlier allows a diagnosis after a single unprovoked seizure when the probability of recurrence over the following ten years reaches 60%. Across studies pooled by the ILAE and International Federation of Clinical Neurophysiology (IFCN), 60% of patients whose EEGs showed IEDs had seizure recurrence after the first unprovoked seizure, against 40% of those whose EEGs did not (Baykan et al., 2025). The presence of IEDs after a first unprovoked seizure is therefore treated as evidence

that the threshold for diagnosis is met, allowing a diagnosis of epilepsy without waiting for a second seizure. Epileptiform discharges are rarely seen in healthy people. Patients with no history of seizures whose EEGs show these discharges often have an underlying acute or progressive cerebral disorder (Sam & So, 2001). In the absence of an unprovoked seizure, the presence of IEDs is not sufficient to diagnose epilepsy, and the risk of seizures developing later is assessed case by case (So, 2010).

Visual review of the EEG remains the gold standard for identifying these discharges, yet that standard is itself unreliable. Experts disagree over whether an individual waveform qualifies as an IED (Jing, Herlopian, et al., 2020), and over-reading is a recognised cause of false-positive epilepsy diagnoses (Benbadis, 2010). The difficulty runs in two directions. IED morphology is not fixed; it varies with the patient's age, the cortical region involved, and the state of vigilance (Aanestad et al., 2020; Fouad et al., 2022). Normal recordings also contain waveforms that resemble epileptiform discharges, since benign variants such as wicket spikes and small sharp spikes, artefacts from eye movement or muscle activity, and fluctuations in the background rhythm are all mistaken for IEDs (Amin et al., 2023; Benbadis & Lin, 2008).

Attempts to distinguish epileptiform discharges from normal sharp transients date back several decades. Maulsby (1971) and Pedley (1980) drew practical guidelines for distinguishing epileptiform discharges from normal sharp transients from clinical experience. These were later consolidated into six morphological criteria (as shown in Figure 1) by the IFCN (Kane et al., 2017) and validated by Kural et al. (2020a) against a video EEG gold standard, with at least five criteria needed to reach a specificity (true negative rate) above 95% to limit misdiagnosis. Adopting the criteria improved the specificity and inter-rater agreement of trainees (Kural, Aydemir, et al., 2022), but produced no improvement in the diagnostic accuracy or inter-rater reliability among experts, and the agreement on the individual criteria among experts was only fair to moderate (Yuan, 2025).

Decades of attempting to formalise IED morphology have not produced a definition that applies uniformly across all discharges, redistributing the ambiguity rather than resolving it.

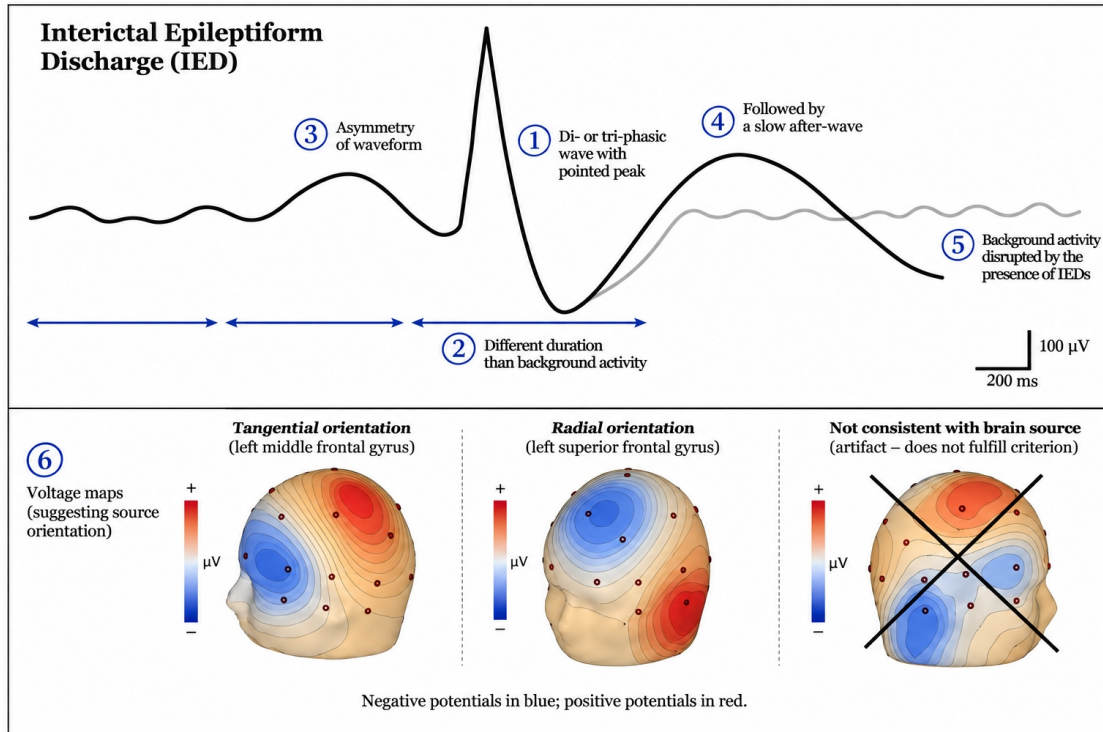


Figure 1

The six IFCN morphological criteria for identifying Interictal Epileptiform Discharges, shown on an illustrated representative discharge. Adapted from Kural et al. (2020a)

Additionally, prolonged monitoring generates hours of recording that experts must review manually. Together, these limitations called for automated IED detection and classification.

Automated IED Detection

Automated detection dates to the early 1970s. The first-ever model flagged waves sharper than the surrounding background (Carrie, 1972). The approach that set the pattern for what followed came from Gotman and Gloor (1976), who decomposed each EEG channel into half-waves characterised by their duration, amplitude, and sharpness relative to the background. These features encode the same properties that a human reader assesses, so the approach reproduces visual review and inherits its limitations. Later detectors added spectral, wavelet-derived and nonlinear descriptors, computing a fixed set of numbers from each candidate and comparing them against thresholds (Abdi-Sargezeh et al., 2024). Nguyen et al. (2026)

extracted 77 such features across morphological, time, frequency, wavelet, and nonlinear domains and classified them using gradient boosting. Their study found that feature sets containing morphological descriptors consistently outperformed those without. The discriminative information, therefore, lies in the shape of the waveform. Fixed thresholds further assume a stable morphology, which IEDs lack (Aanestad et al., 2020; Fouad et al., 2022). Gotman and Wang (1991) responded by classifying the EEG into vigilance states and setting a separate threshold for each, which partitions the variation without modelling it.

Template-based Methods

Template-based methods compare waveforms against a template or reference. A single template cannot capture the heterogeneity of IEDs, and broadening it enough to do so also admits mimics, keeping the false-positive rate high. Lodder et al. (2013) addressed this by building a database of over 2000 templates learnt from training data, covering the most common spike morphologies, so that no single reference had to represent all of them. The templates are whole discharges, not components of an IED that combine. Waveforms are matched to stored templates, and a morphology unlike anything in the database cannot be expressed in terms of the templates already held. Enlarging the database therefore adds more examples without describing how discharges differ from one another.

Template matching can also generate candidates. Every comparison against a template produces a similarity score, and Lodder et al. (2013) used those scores to rank detections so that a reviewer could examine the closest matches first. Lodder and van Putten (2014) extended this into a system that adapted its rankings as the reviewer confirmed detections. The template comparison here is not deciding which candidates are IEDs. It generates candidates and orders them for experts to review.

Jing et al. (2016) identified a different source of failure in the template-matching itself, finding that rigid point-wise comparison rejected waveforms whose morphology matched a template but whose features fell at different latencies. They aligned the waveforms using dynamic time warping before comparing them to the template. Dynamic time warping (DTW) is an

algorithm commonly used in time-series analysis to measure the similarity between temporal sequences that may vary in length, speed, or time. After the alignment, the candidate is still represented by its similarity to a stored template. How much warping the alignment required is absorbed into the optimisation and does not appear in the result, so two candidates that match a template equally well may have required very different amounts of warping. DTW provides no space in which those warpings can be averaged or compared, and the timing differences are therefore never examined as a possible source of discrimination. After the alignment, the candidate is still represented by its similarity to a stored template.

Template-matching candidate-generation methods were developed to speed up expert annotation. A reviewer inspects the candidates that the system flags rather than reading the full recording, and Jing et al. (2016) reported a roughly 70% reduction in annotation time. A limitation here is that a discharge that is unlike anything in the database is dropped as a candidate and never reaches the reviewer. More templates increase the chance of a close match, but since each one is a whole discharge, coverage extends only as far as the stored examples. The true positive rate, or recall, measures how many of the true positives a system correctly identifies. Recall at this first stage therefore sets a ceiling on the whole system, since an IED dropped during candidate generation cannot be recovered downstream. This applies to any staged detector: the generating layer has to be permissive enough to keep almost every discharge, and the cost of that leniency is more candidates for the later stages to sort through.

Zacharaki et al. (2016) classified the waveform as a whole, embedding full candidate waveforms in a low-dimensional space using locality-preserving projections and classifying them with support vector machines. They showed that a low-dimensional representation of the shape carries more information than scalar descriptors. This embedding, however, treats each waveform as a point in Euclidean space, imposing no smoothness on the curve and no separation of timing from form. Hence, latency differences register as differences in shape.

Deep Learning Methods

Deep learning removed the need to specify a representation in advance. Jing, Sun, et al. (2020) trained SpikeNet, a convolutional neural network, on 9,571 scalp recordings annotated by eight fellowship-trained neurophysiologists, and reported detection performance that matches expert level. Tjepkema-Cloostermans et al. (2025) compared a deep network with seven experts and matched them on sensitivity (true positive rate; also known as recall) at 99.0% specificity. Alkofer et al. (2025) trained another deep neural network, Cyclops, on several thousand patients to detect discharges from a reduced electrode set.

Performance drops on recordings from other sources. On an external set, Cyclops fell from an internal AUROC ¹ of 0.950 to a median of 0.846, with values as low as 0.692 (Alkofer et al., 2025). Nhu et al. (2023) found the failure to be directional: models trained on their private recordings generalised to the public Temple University Hospital EEG Events Corpus (TUEV), while models trained on TUEV did not generalise to the private recordings, which they attributed to differences in class distribution between the datasets. The problem is not specific to deep learning, since Nguyen et al. (2026) report cross-dataset sensitivity falling to 50.6% for a feature-based classifier.

These evaluations score detections against transients marked by experts. Kural, Jing, et al. (2022) instead confirmed the diagnosis by video-EEG and scored each system on whether it separated patients with epilepsy from patients with non-epileptic events. Networks such as Persyst, Encevis and SpikeNet reached usable sensitivities with specificities below those of the experts, and expert review of the detections was required to raise specificity to an acceptable level.

These systems require large annotated datasets and give no description of the waveform properties behind a detection. Clinicians cannot check the reasoning behind a flagged transient, and automated detection models remain rarely used in practice (Cook et al., 2026; Reus et al., 2023). Researchers have built interpretable models for this task. For instance, Tang et al. (2025)

¹AUROC - Area under the Receiver Operating Characteristic curve. Measures a binary classifier's ability to discriminate between classes (0.5 = random chance, 1.0 = perfect performance).

introduced ProtoEEG-kNN, which adds a retrieval step over the deep network architecture: a candidate is classified by retrieving the most similar recordings from the training set and displaying them, reaching state-of-the-art accuracy with explanations that experts preferred over existing approaches.

Identifying a discharge is a qualitative visual skill acquired through training. Halford et al. (2018) had 35 neurologists mark discharges in routine EEG segments and found only fair agreement between them, with performance varying by practice setting, fellowship training and board certification. Tjepkema-Cloostermans et al. (2025) report pairwise sensitivities between experts from 20.7% to 86.4% on the same recordings, and the IFCN criteria intended to standardise these judgements did not improve agreement among experts (Yuan, 2025). Annotations therefore record how a particular set of readers was trained to identify IEDs. A detector trained and evaluated on those annotations reproduces the readers' judgement and inherits its inconsistency.

Functional Data Analysis

Across these approaches, the classifier receives a fixed set of numbers computed from the candidate waves: quantities measured at chosen points, a distance from stored examples, or values learnt from labelled data. The waveform itself does not reach the classifier, and its shape between the points is either discarded or not measured. Functional Data Analysis (FDA) is a statistical modelling framework that treats each observation as a curve instead of a vector of isolated data points (Ramsay & Silverman, 2005). A signal sampled at discrete points is represented as a continuous function, which can then be smoothed, differentiated, aligned with other curves, and compared in terms of shape.

Two choices precede any functional analysis. The first is the domain on which the curve is defined. In the time domain, the signal, $x(t)$, is the voltage recorded at each instant. This domain holds the duration of events, their sequence, and their positions in a recording. In the frequency domain, the curve is the power spectrum $|X(f)|^2$, which shows how much energy the signal carries at each frequency over the entire analysed segment. Moving from the time domain to the

frequency domain requires a Fourier transform, which yields the complex coefficients $X(f)$, followed by their squared modulus. A third possibility retains both axes. A time-frequency representation such as a spectrogram assigns power to each frequency at each instant, but the result is a surface over a two-dimensional domain rather than a curve.

The second choice is the set of basis functions used to represent the curve. A curve is approximated as a weighted sum of K basis functions,

$$\hat{x}(t) = \sum_{k=1}^K c_k \phi_k(t) \quad (1)$$

where the ϕ_k are the basis functions, and c_k are the coefficients fitted to the observed samples, and K is the number of basis functions. Choosing a smaller K than the number of samples smooths the curve, reducing noise. The fitted function is defined at every point in its domain rather than only where the signal was sampled. This lets you take derivatives, align curves and compare shapes.

A Fourier basis represents a curve as a weighted sum of sine and cosine waves at multiples of a base frequency. Each wave extends across the entire curve, so a coefficient describes how much of one frequency is present in the whole curve. This suits periodic curves. Representing a feature confined to a small region is less efficient, since it requires several frequencies combining so that they sum to the shape of the feature in that region and sum to zero everywhere else. A curve that is not periodic is also fitted poorly at its two ends, where the basis constrains it to join back to itself. A wavelet basis is generated from a single function, the mother wavelet, by shifting it along the curve and by stretching or compressing it, the amount of stretching being the scale. Each basis function is therefore identified by a position and a scale, and its coefficient measures how closely the curve resembles that wavelet at that place and width. Because every coefficient carries both indices, the expansion returns a two-dimensional array rather than a curve. B-splines are short polynomials joined at points called knots, with each polynomial covering only a short section of the curve. A coefficient gives the weight of one such polynomial, so it affects only that section, and a change in one part of the curve does not alter the fit elsewhere. The order of the polynomial controls how smoothly the segments join at the knots. A higher order makes the transition between segments more gradual, so the fitted curve has no

kinks² and its curvature is defined at every point. The number and placement of knots have to be chosen, with too many following noise in the data and too few smoothing away features that matter. Wavelets and b-splines are both localised, and both therefore suit curves that are irregular or non-periodic. The domain and the basis are chosen separately, and applications of FDA to epilepsy EEG have so far differed in the choice of domain.

Xie and Lawniczak (2019) applied FDA to epilepsy EEG for two tasks: separating recordings from healthy volunteers and from patients with epilepsy, and detecting seizures. Each recording segment was transformed into its Fourier power spectrum, then expanded in a B-spline basis and passed through Functional Principal Component Analysis (FPCA). The functional object is therefore power as a function of frequency, describing regularities that hold across a whole segment. A later study by Xie (2021) used a wavelet power spectrum instead of a Fourier power spectrum, followed by B-spline expansion and FPCA, for the same two tasks and for separating segments recorded during a seizure from segments recorded between seizures. Every comparison in both studies was made at the level of the segment, with no reference to individual discharges.

Xie and Lawniczak (2019) argue that raw EEG signals are poor functional objects, and that transforming to a power spectrum is what makes FPCA meaningful. This holds for the tasks they set out to do, although Yi et al. (2022) found a time-domain functional model to perform comparably to a transformed one on a cognitive classification task, so the requirement is not general. Choosing the domain requires making a distinction between a process and an event. A seizure is a process: rhythmic activity the brain enters and sustains for many seconds, raising power in particular frequency bands well above background for the whole segment analysed. A power spectrum represents this, and a raw trace does not. An IED is an event lasting 20 to 200 ms. Within a segment of several seconds, it accounts for a small fraction of the total energy, and that energy spreads across the spectrum instead of appearing at a single frequency. The squared modulus also discards the phase, so the order of the sharp deflection and the slow after-wave that

²A point where a curve changes direction abruptly, corresponding to a discontinuity in its first derivative.

follows it cannot be recovered, and duration, which separates a spike from a sharp wave, cannot be read from a spectrum at all.

Scope of this Study

This study uses FDA to detect IEDs in scalp EEG, applying B-spline expansions to candidate transients in the time domain. Registration aligns each curve to a template, separating variation in shape from variation in timing. Functional principal component analysis reduces each to a set of components. The components are estimated from the training data and combined, so an unfamiliar morphology is expressed through shared modes of variation. Each component is a curve on the same time axis as the waveform and can be plotted, showing directly which part of the waveform varies along it.

We first evaluate the representation as a classifier, separating IEDs from non-epileptic paroxysmal events, and testing shape and timing as sources of discrimination. We then place the classifier within a staged detector, where earlier layers identify candidate transients across a recording and pass them to the classifier. We measure performance on continuous recordings under cross-validation grouped by recording, at an operating point fixed by a false-positive budget set in advance. A second dataset tests whether the number of detections separates recordings in which experts identified discharges from those in which they did not.

Methodology

Dataset

We used a secondary dataset collected by Kural et al. (2020b) from 100 patients who underwent long-term video-EEG monitoring. Video-EEG monitoring records the EEG and a synchronised video of the patient so electrical events can be traced to patient movement and behaviour. It was recorded using the standardised 25-electrode array at 500 Hz. Fifty-four patients had epilepsy, and 46 had non-epileptic paroxysmal events. Recordings comprised one interictal epoch of 10 to 20 seconds, each containing a single marked sharp transient (IEDs in the epileptic recordings and non-epileptic paroxysmal events in non-epileptic recordings). IEDs were further divided by how readily they were identified as IEDs, giving clear and unclear positives. Of

the 54 IEDs, 32 were clear positives and 22 were unclear.

Functional Representation Study (Exploratory)

This initial study tests whether the functional representation separates IEDs from non-epileptic paroxysmal events before it is placed within a detector, and whether the discriminative information is carried by the registered curves or by the warping that aligns them. We cut a two-second window around each annotation, centred on the annotated timestamp. We took the window from a single representative channel, selected as the channel with the largest peak-to-peak amplitude within it, and divided it by the recording’s background amplitude. We measured background amplitude in two steps: we computed the median absolute deviation for each channel across the whole preprocessed recording, then took the median of those values as the recording’s background level.

Each window was expanded in a B-spline basis ($\hat{x}(t) = \sum_{k=1}^K c_k B_k(t)$) (shown in Figure 2a) and aligned using Fisher-Rao elastic registration, a method that time-warps each curve onto a common template. The template is a single Karcher mean³ computed over all training windows. Registration decomposes each curve into two objects: the registered curve, which is the waveform once aligned to the template (the amplitude component), and the warping function (the phase component), which records how much stretching the alignment required. Figure 2b shows how curves align after applying registration. We applied functional principal component analysis to the windows (shown in Figure 2c), and classified resulting functional principal components with logistic regression.

We compared four conditions: FDA amplitude condition, FDA phase condition, FDA amplitude + phase conditions, and a scalar baseline measured as the peak-to-peak amplitude of the spike. Comparing the amplitude and combined conditions asks whether adding the phase

³The generalization of the average to curved geometric spaces, calculated here as the single reference curve that minimizes total shape distance across all training windows without blurring their features.

component improves discrimination. For this, we used two one-sided tests (TOST)⁴ and set the margin at ± 0.05 AUROC, consistent with previous ROC-equivalence studies. We then compared the amplitude condition against the scalar baseline using a paired bootstrap⁵ over recordings with 2000 draws.

We used nested cross-validation with five inner and five outer folds over ten repeats so that the reported performance is unbiased by parameter selection. Parameters were chosen in the inner folds using a one-standard-error rule, and performance was measured on the outer folds. We fitted registration and functional principal component analysis within each training fold. The nested cross-validation used 30 basis functions, five components, a registration penalty of 0.5.

Staged Detection

Detection follows a layered architecture. Earlier layers identify candidate transients and pass them to the classifier, which scores each one; a final layer then combines that score with spatial features. Because a discharge lost at the candidate stage cannot be recovered downstream, the earlier layers are set to favour recall. The following subsections describe the layers. Figure 3 summarises the full pipeline.

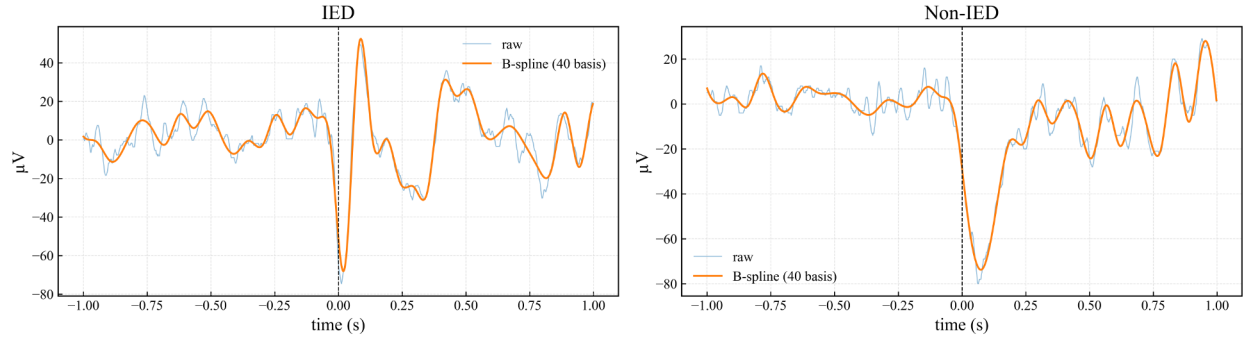
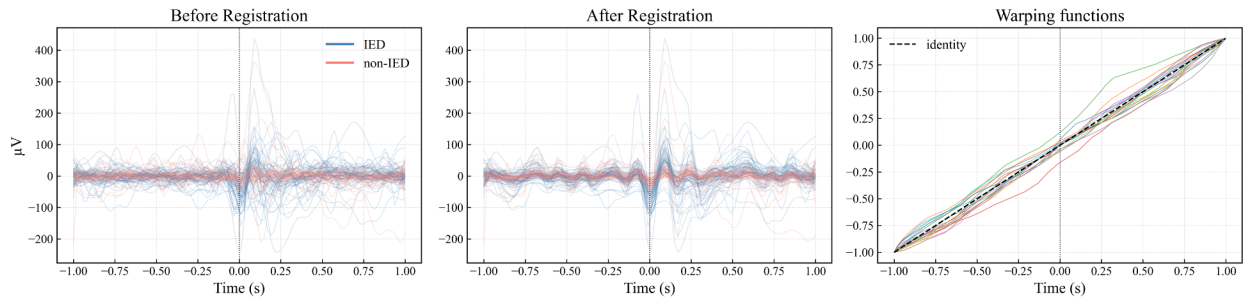
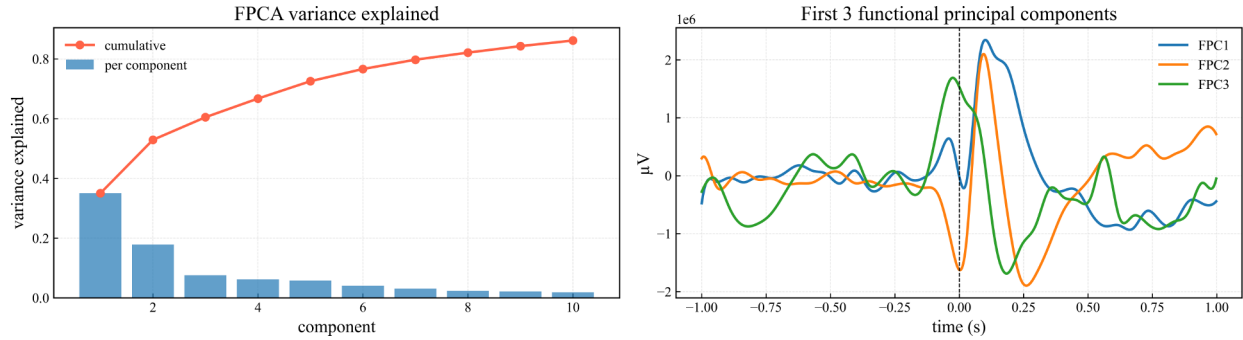
Preprocessing

Signal preprocessing included:

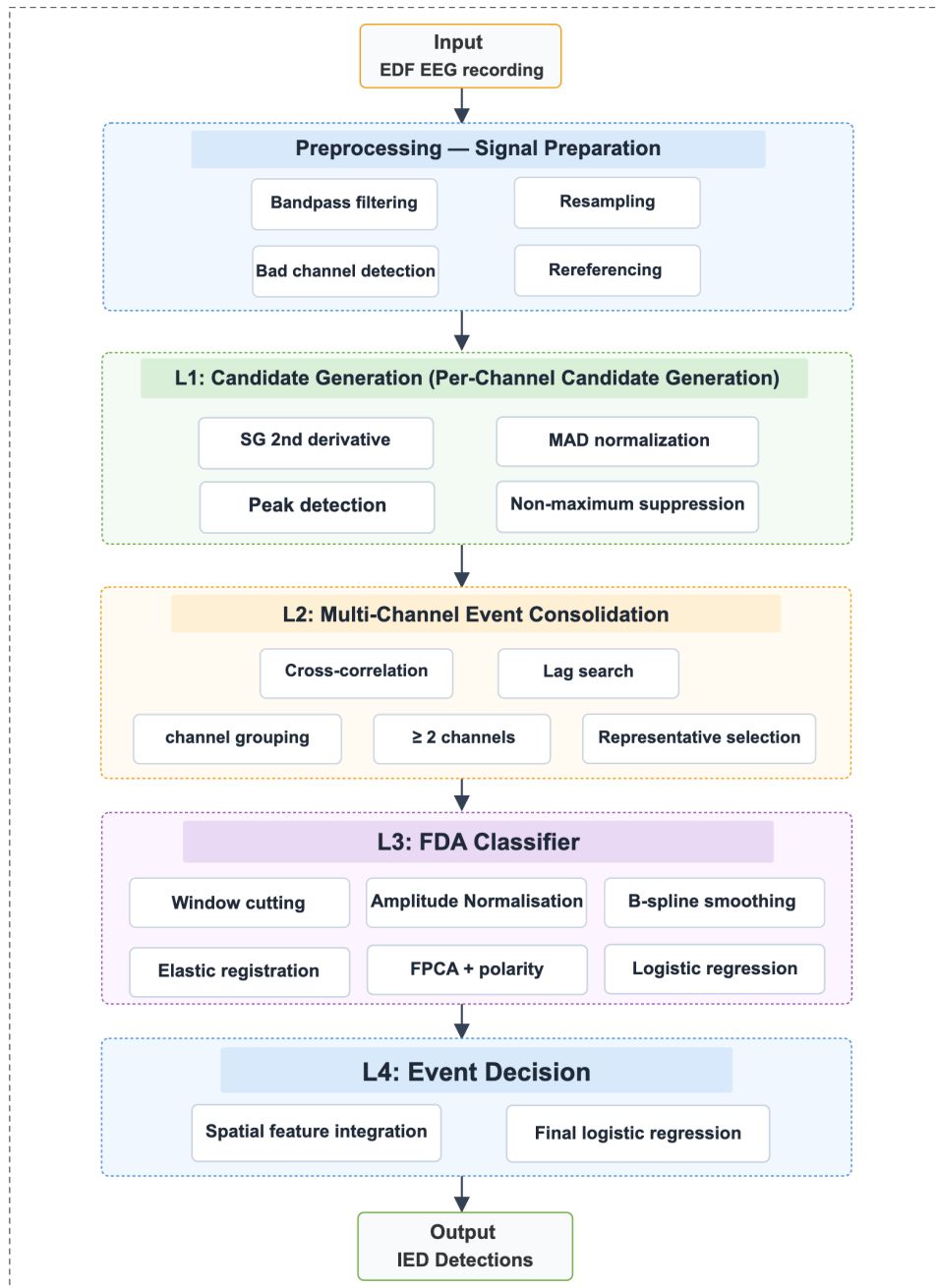
- **Channel matching:** Channel names were normalised and then matched to the 25 electrodes.
- **Frequency filtering:** We applied a zero-phase bandpass filter from 0.5 to 45 Hz. The lower edge retains the slow after-wave, which forms part of the discharge; the upper edge

⁴Two one-sided tests (TOST) assess whether the difference between two conditions is sufficiently small to be considered practically equivalent. It does this by testing whether the difference lies within a prespecified equivalence margin in both directions.

⁵Paired bootstrap resampling estimates uncertainty by repeatedly resampling observations with replacement while keeping the corresponding measurements from the two conditions together within each resample. This preserves the dependence between paired observations and allows the distribution of their differences to be estimated.

(a) *B-spline Expansion for IED and Non-IED*(b) *Fisher Rao Elastic Registration*(c) *Functional Principal Component Analysis***Figure 2**

Panel a shows the B-spline expansion with 40 basis functions of a two second window containing an IED (on the left) and a 2 second window containing a mimic (on the right). Panel b shows registration on the training windows. Left: the windows before alignment. Centre: the windows after warping onto the Karcher mean template. Right: the warping functions that produced the alignment. Panel c shows the functional principal components of the registered windows. Left: variance explained by each component and cumulatively. Right: the first three components.

**Figure 3**

The detection pipeline summary. Recordings are preprocessed, candidate transients are generated per channel, consolidated into multi-channel events, scored by the functional classifier on the representative channel, and combined with spatial features to produce a final score for each event.

excludes both 50 and 60 Hz mains interference, so a notch filter was not required. We used a zero-phase filter to preserve the timing and shape of each transient.

- **Resampling:** signals were resampled to 250 Hz. This retains sufficient resolution for transients of 20 to 70 ms while halving subsequent computation cost.
- **Bad channel detection:** we applied two thresholds to the median absolute deviation of each channel, expressed as a multiple of the median across channels. Channels exceeding three times the median were excluded from the average reference so noisy channels did not contaminate it, but their data was retained for analysis. Channels exceeding ten times the median, or with a peak-to-peak amplitude below $0.5\mu V$, were treated as unusable and replaced by spherical spline interpolation from neighbouring electrodes.
- **Rereferencing:** We rereferenced signals to the average across all good channels.

Layer 1: Candidate Generation

This layer generates candidate transients to feed into the FDA classifier. To do this, we first smoothed each channel with a Savitzky-Golay filter⁶ over a 42 ms window at the 3rd order. We then took the 2nd derivative of the smoothed signal, which is large where the signal turns sharply and near zero on a smooth background.

We then divide the absolute curvature by that channel's median absolute deviation (MAD), so the measure reflects how sharp a transient is relative to the channel's usual sharpness rather than its absolute amplitude. Peaks exceeding 6 MAD become candidates, with 150 ms non-maximum suppression so that one biphasic complex produces a single candidate per channel. Candidates are generated from all channels except interpolated ones. Candidate transients across channels from the same discharge must be grouped into a single event before being fed into the classifier.

⁶A digital smoothing filter that removes noise from data while preserving the high-frequency features and sharp peaks of the original signal

Layer 2: Event Consolidation

This layer groups candidates on different channels arising from the same discharge into a single event. To do this, we compare candidates within 50 ms of each other by cross-correlation (threshold 0.7) over a window of ± 150 ms, wide enough to contain the discharge and its slow after-wave. We allowed a lag search of 30 ms to account for differences in signal arrival time between electrodes. We normalised the correlation so a diminished copy on a distant channel still matches. We also used the absolute value of the correlation to include the polarity-reversed side of the dipole. We test candidates on every pair of channels rather than comparing each to a single reference channel, since a candidate at the edge of a discharge's field need not resemble the sharpest channel. An event is a group of candidates connected by those links: if A links to B and B links to C, all three belong to one event, whether or not A correlates with C directly. Lastly, we reject single-channel events because a discharge arises from a patch of synchronously firing cortex and produces a field across several electrodes. A sharp transient confined to one channel is more consistent with an artefact. Finally, we select a representative channel with the highest peak-to-peak amplitude to pass to the classifier in layer 3.

Layer 3: FDA Classification

This layer includes a logistic regression classifier that scores the representative channel's two-second window on how likely it is to contain an IED. To do this, we first centre the window on the largest deviation of the smoothed signal from its local median within ± 150 ms of the candidate, since no annotation is available at detection time. We divide each window by the recording's background amplitude, apply the B-spline expansion (70 basis functions, 4th order, resampled to a 100-point grid), and apply registration and FPCA. The Logistic regression uses the first 24 functional principal components and the polarity (whether its central deflection is negative or positive) of the transient as features. The parameters selected in the functional representation study did not transfer. Far more candidates, with more heterogeneity, reach this layer; therefore, separating them required a finer representation, and we used a higher basis count and component count. The training pool for this layer is roughly 98.5% negative. We addressed this imbalance

when fitting the logistic regression, by giving each window a weight inversely proportional to the size of its class. A positive window therefore counts about 35 times a negative one, so positives and negatives contribute equally to the fit while every window is still used. We did not resample the training data, and we fitted the registration and FPCA on the whole pool without reference to the labels, so the representation remains unsupervised. This layer returns a probability score for each event.

Layer 4: Event Decision

The classifier in the previous layer sees discharge from a single representative channel, so it discards the shape of the electric field across the scalp. In this layer, we introduce a second logistic regression that combines the layer 3 classification score with two spatial features to produce the final score for each event. The first feature is the number of channels the discharge spreads over, and the second is a dipole measure. The dipole measure records how far apart the positive and negative member electrodes sit relative to the width of the event, since a discharge is generated by a dipolar source and should therefore produce two spatially separated groups.

Training and Validation

We trained the deployed model on all 100 recordings. For reporting, we estimated performance by cross-validation on the same 100. We ran candidate generation and consolidation once over every recording before splitting, producing a pool of roughly 5,500 events. Each event carries its two-second window, the recording it came from, and a label: positive if the recording is from a patient with epilepsy and the event falls within ± 100 ms of the annotated transient, negative otherwise. We then split events into five folds grouped by recording so every event from a recording falls in the same fold. We hold out each recording exactly once. Within each fold, we fitted the classification layer from scratch on the training windows. This included the registration, FPCA, and logistic regression. We then registered and scored the held-out windows. We pooled the resulting out-of-fold scores to obtain one score per event. We fitted the final decision layer within the same folds, using the out-of-fold classification score, the number of member channels, and the dipole measure. We pooled its out-of-fold scores in the same way, and we computed all

reported performance on these.

We measured performance of layers one and two by recall. A discharge dropped here cannot be recovered downstream, so recall at this stage sets a ceiling on the whole detector. We then measured detection performance with an FROC (Free-response Receiver Operating Characteristic) curve, which plots sensitivity against false positives per minute across all regression thresholds. This lets us choose an operating point for the classifier. We count a true positive if any event within ± 100 ms of its annotation clears the threshold, and we count every other event above the threshold as a false positive. Roughly 98.5% of the candidates generated in our pipeline are negatives, so a classifier that rejects every candidate, including all IEDs, would still reach 98.5% accuracy. We therefore do not report accuracy, since it is determined almost entirely by the true negatives and says nothing about how many discharges were found. We instead report PR-AUC, the area under the precision-recall curve, which measures how many of the flagged candidates are IEDs against how many of the IEDs present were flagged. We also report the median distance between each detection and its annotation as a measure of localisation error.

External Testing

We used a second dataset of 50 clinical recordings, each roughly 20 minutes long, for external testing. 25 were confirmatory, in which experts had identified IEDs, and 25 were non-confirmatory, in which experts found no IEDs. Among the non-confirmatory recordings, 11 came from people with epilepsy, too few to analyse as a separate subgroup. No annotations were provided within the recordings, so we had only recording-level labels. The data provider ran the analysis with the scripts we provided and returned per-recording and per-event outputs. Because these recordings contain no marked discharges, we could not compute sensitivity or localisation. We therefore compared the number of detections per recording between the confirmatory and non-confirmatory groups.

Implementation

The pipeline is implemented in Python. We loaded and preprocessed recordings with MNE (v1.8.0), which also provided the template electrode coordinates. Basis expansion and functional principal component analysis used scikit-fda (v0.9.1) (Ramos-Carreño et al., 2024), and Fisher-Rao elastic registration used fdasrsf (v2.6.1). We fitted classifiers with scikit-learn (v1.6.1). We saved the fitted models with joblib (v1.5.3).

Results

Functional Representation Results

This initial study aimed to determine whether functional representation of the signal outperforms a scalar baseline (the raw peak-to-peak amplitude of the spike) and whether the discriminative information lies in the shape of the waveform or the timing of its features. Table 1 reports classification performance across the three FDA conditions and the baseline.

Table 1

Classification performance across feature conditions

Condition	AUROC	PR-AUC	Sensitivity clear +ve	Sensitivity unclear +ve	Specificity
FDA amplitude	0.899	0.904	0.807	0.706	0.860
FDA phase	0.615	0.602	0.532	0.622	0.551
FDA combined	0.906	0.918	0.786	0.723	0.833
Scalar baseline	0.746	0.704	0.625	0.520	0.787

Note. Bold marks the best value in each column. The AUROCs and the PR-AUCs of the FDA amplitude and combined conditions are statistically equivalent (see text).

We calculated all AUROCs and PR-AUCs with 95% confidence intervals. The FDA amplitude condition (AUROC 0.899; range 0.831 to 0.955) and FDA combined condition (AUROC 0.906; range 0.842 to 0.960) showed equivalent discrimination. The AUROC difference

was $=0.007$ (90% CI: $=0.032$ to 0.016 ; TOST $p = 0.002$), and fell within the prespecified AUROC equivalence margin of ± 0.05 , indicating that adding the phase component to the amplitude component did not provide a practically meaningful improvement in discrimination. This is consistent with the lower performance of the phase condition alone, which had an AUROC of 0.615 (0.498 to 0.729), with its confidence interval spanning the chance level. The FDA amplitude condition achieved the highest sensitivity on clear positives and the highest specificity, and sensitivity was lower on unclear positives than on clear positives in every condition except phase. The PR-AUCs were 0.904 (0.811 to 0.969) for the amplitude condition, 0.918 (0.846 to 0.968) for the combined condition, and 0.602 (0.486 to 0.757) for phase alone. The chance level for the PR-AUC is 0.54 so the FDA amplitude condition is 1.67 times the chance level. The PR-AUC difference between the FDA amplitude and FDA combined conditions was -0.014 (95% CI -0.052 to $+0.019$, two-sided $p = 0.486$), so the two do not differ on this metric either.

The scalar baseline reached an AUROC of 0.746 (0.638 to 0.840). It reached a specificity of 0.787 , close to that of the FDA amplitude and FDA combined conditions. Its sensitivity, however, was the lowest of the four on both clear positives (0.625) and unclear positives (0.520), the latter barely above half, so the baseline misses roughly two in five clear IEDs and half of the unclear IEDs. A paired bootstrap over recordings gave an AUROC difference of $+0.153$ (95% CI $+0.051$ to $+0.263$, two-sided $p = 0.001$) in favour of the FDA amplitude condition. The PR-AUC difference was $+0.199$ (95% CI $+0.071$ to $+0.311$, $p < 0.001$), also in favour of the FDA amplitude condition. Of the threshold-dependent metrics, only sensitivity on unclear positives resolved, at $+0.219$ (95% CI $+0.031$ to $+0.406$, $p = 0.037$). Specificity differed by $+0.130$ (95% CI $=0.020$ to $+0.275$, $p = 0.112$) and sensitivity on clear positives by $+0.182$ (95% CI $=0.080$ to $+0.469$, $p = 0.261$). All five differences favour the functional representation, and the confidence intervals on the threshold-dependent metrics span roughly 0.2 to 0.3 .

Detection Results

We used the FDA amplitude condition in the classifier for the detection pipeline. The detection pipeline was split into 4 layers, with the earlier layers generating candidates and the

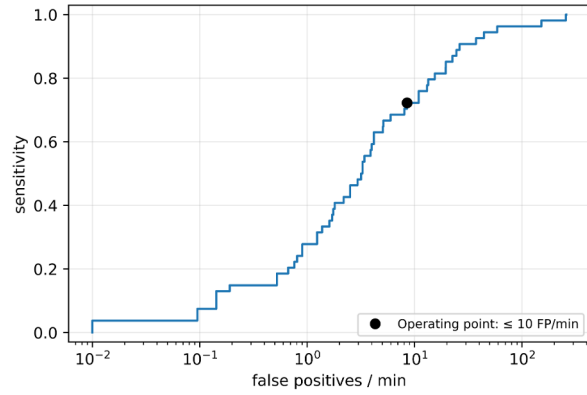
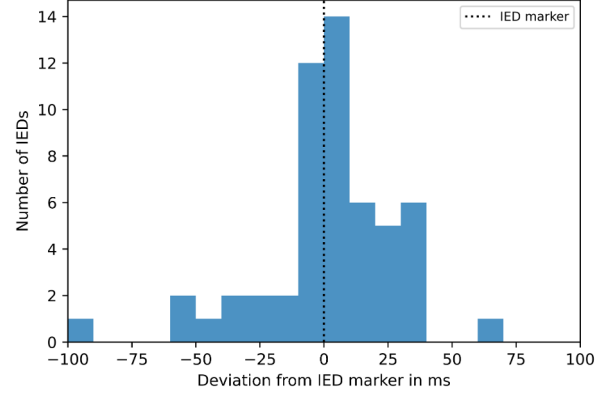
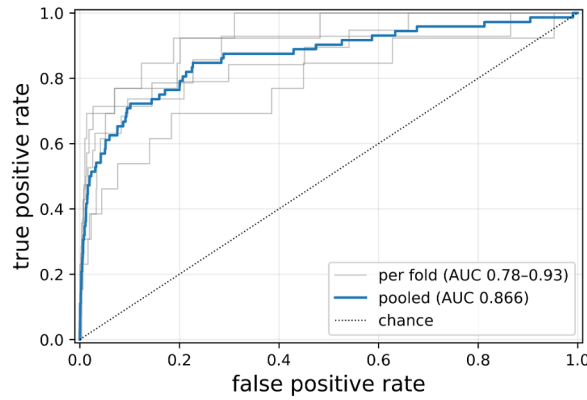
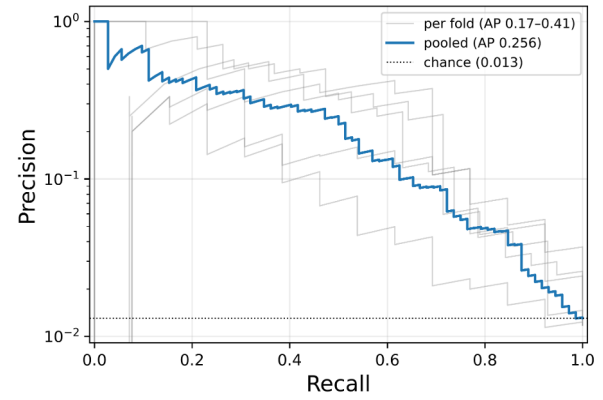
later layers discriminating between IEDs and non-IEDs. The recall of both layers one and two was 1.0, which means that all IEDs were identified as candidates. We could not measure consolidation performance because the dataset does not record which channels a discharge appeared on. From these layers, we generated 5519 candidates from all 100 recordings.

Layer three, which used only the functional principal components and polarity features, achieved an AUROC of 0.827 (95% confidence interval 0.760 to 0.893) and a PR-AUC of 0.126 (approximately 10.5 times the chance level). This was with a false-positive budget of ≤ 10 , giving a sensitivity of 0.56. At layer four, the pipeline reached a sensitivity of 0.72, detecting 39 of the 54 IEDs with 179 false positives over 21.1 minutes of recording (8.5 per minute). This was at the same false-positive budget of ≤ 10 per minute. Table 2 shows the performance comparison of layers three and four. Figure 4a shows the FROC curve. ROC-AUC was 0.866 (95% confidence interval: 0.81 to 0.92), with individual fold estimates ranging from 0.78 to 0.94. PR-AUC was 0.256, roughly twenty times the chance level of 0.013. Figures 4c and 4d show both the ROC and PR curves. The median localisation error was 12 ms across the 39 detected IEDs (localisation error distribution shown in Figure 4b).

Table 2

Performance comparison of layers three and four

Metric	Layer 3	Layer 4
Features	FPCs + polarity	L3 scores + number of channels + dipole
AUROC	0.827	0.866
PR-AUC	0.126	0.256
Sensitivity	0.56	0.72
False positives	204	179
IEDs detected (hits)	30/54	39/54
IEDs missed (misses)	24	15
False positives per minute	9.7	8.5

(a) *FROC Curve*(b) *Localisation Error*(c) *Receiver Operating Characteristic (ROC)*(d) *Precision-recall Curve*

Curve

Figure 4

Panel a: cross-validated FROC curve, plotting sensitivity against false positives per minute over out-of-fold scores. The operating point is at a budget of ≤ 10 false positives per minute, giving a sensitivity of 0.72. **Panel b:** Localisation error, calculated as deviation from IED annotation in ms. The median localisation error is 12 ms. **Panel c:** Out-of-fold ROC over candidate windows. Individual folds shown in grey (0.78 - 0.94), pooled shown in blue (0.87). **Panel d:** Precision-recall curve calculated on out-of-fold scores. The dashed line marks the chance level of 0.013, given by the proportion of positive events. PR-AUC was 0.26.

External Testing

We conducted external testing to assess whether the detection model would generalise well to real-world clinical data. We compared the two groups on two statistics: detections per minute (events clearing the threshold, divided by recording length) and mean top-10 (the mean of the ten highest event scores in a recording, which does not depend on a threshold).

The confirmatory group had a median of 15.21 detections per minute, and the non-confirmatory group had a median of 11.77. Ranking recordings by this rate separated the groups with an AUROC of 0.582 (0.419 to 0.741) and a one-sided Mann-Whitney p-value of 0.161. The interval spans 0.5, so the rate does not separate the groups.

Ranking recordings by mean top-10 gave an AUROC of 0.715 (0.563 to 0.858), with a one-sided Mann-Whitney p-value of 0.005. This separates the groups, so they differ in how strongly the detector responded to their best candidates rather than in how many events cleared the threshold. This is consistent with a separate analysis we performed across false-positive budgets, in which the ratio of median detection rates between the groups was 2.10 at 1 false positive per minute and 1.14 at 5 false positives per minute.

Discussion

This study asked four questions: whether a functional representation of candidate transients separates IEDs from non-epileptic paroxysmal events, whether the discriminative information lies in the shape of the waveform or in the timing of its features, whether that representation supports detection in continuous recording, and whether the number of detections can separate recordings in which experts found IEDs from recordings in which they did not.

Our functional representation study showed that the functional representation ranked candidates better than peak-to-peak amplitude using paired bootstrap resampling. The AUROC difference was +0.153. Specificity and sensitivity on both clear and unclear positives also favoured the functional representation. The scalar baseline reaches a specificity of 0.787, which is acceptable; however, it reaches a sensitivity of only 0.520 on unclear IEDs and misses almost half of them, so it performs poorly on ambiguous discharges. This result is consistent with Zacharaki

et al. (2016), who found that a low-dimensional representation of shape carries more information than scalar descriptors.

Our second question asked whether the discriminative information lies in the waveform's shape or in the timing of its features. Two results point in the same direction. The phase condition, which uses only the warping functions that align each curve to the template, reached an AUROC of 0.615 with an interval spanning chance. An equivalence test then showed that adding the phase component to the amplitude component changed discrimination by less than the prespecified margin of ± 0.05 . Together these suggest that the warping functions don't carry discriminative information, and that the shape of a transient separates discharges from mimics while the timing of its internal features does not. Jing et al. (2016) and Zacharaki et al. (2016) both set aside the timing variation in their candidates, one by absorbing it into the warping optimisation and the other by leaving it entangled with shape. Our results suggest that neither choice cost them much discriminative information. These results, along with the previous question's results, are consistent with Nguyen et al. (2026)'s finding that morphological features outperform other features.

These results do not show that timing is irrelevant to identifying a discharge. They show that the amount of stretching or compression needed to align transients to a template does not carry discriminatory information. The phase condition did not test other temporal properties. Each window was centred on its annotated timestamp, so the transient's position in the recording was fixed before registration ever ran. Duration and the asymmetry between a sharp ascending phase and a slower decay are temporal properties too, and both survive alignment, so they sit inside the amplitude component rather than the phase component.

Our third question asked whether the functional representation supports detection in continuous recording. From layer three alone, we got an AUROC of 0.827 and a sensitivity of 0.56 at a budget of 10 false positives per minute. The exploratory functional representation study reached an AUROC of 0.899 on the same representation but against 46 non-IEDs, which was 46% of the entire sample size. In the detection pipeline, however, the negative class is roughly 98.5%.

The imbalance does not affect AUROC much; however, PR-AUC drops from 0.904 in the exploratory study to 0.126 in the detection pipeline, meaning the classifier returns far more false positives per true detection. The parameters used in the functional representation study did not transfer to the detection pipeline. The candidates were more heterogeneous and hence required a finer resolution. Layer three therefore used 70 basis functions and 24 functional principal components as against 30 basis functions and 5 functional principal components in the functional representation study. Functional representation therefore detects discharges in continuous recording, though at 9.7 false positives per minute it flags far more non-IEDs than IEDs. Part of that may reflect the training data rather than the representation, since only 54 positives were available against several thousand negatives.

Layer three scores a single channel, so a transient whose shape resembles a discharge cannot be rejected on morphology alone. Layer four therefore adds two spatial features, the number of channels the event spreads over and a dipole measure. Adding them raised sensitivity at the same false positive budget from 0.56 to 0.72 and roughly doubled the PR-AUC, while the AUROC moved only from 0.827 to 0.866. Spatial features therefore change which candidates end up at the top of the ranking without much changing the overall ordering, which is why AUROC moves so little while sensitivity and PR-AUC rise. Those candidates are the ones whose shape resembles a discharge on a single channel, and what separates them is the field: a discharge arises from a dipolar source and appears across several electrodes, while an artefact with a similar shape need not. This corresponds to the sixth IFCN criterion, the distribution of potentials across the scalp (Kane et al., 2017). Roughly a third of the sensitivity at the operating point therefore comes from spatial information.

Our fourth and last question asked whether the number of detections separates confirmatory recordings (recordings in which experts found IEDs) from non-confirmatory recordings (recordings in which experts did not find IEDs). The median detection rate was 15.21 per minute in the confirmatory group and 11.77 in the non-confirmatory group. Classifying recordings by detection rate gave an AUROC of 0.582 (0.419 to 0.741), so the rate does not

separate the two groups. Classifying by mean top-10 score gave an AUROC of 0.715 (0.563 to 0.858), suggesting that the higher-scoring candidates separate the two groups rather than the number of events clearing a threshold. The sweep across false-positive budgets shows the same pattern. The ratio of median detection rates between the groups was 2.10 at one false positive per minute and 1.14 at five false positives per minute, so the separation is present when we count only the strongest candidates. The number of detections therefore does not separate the two groups on this dataset, though the highest-scoring candidates show some separation between recordings.

The confirmatory recordings contain discharges, but they were never marked, so we could not inspect the highest-scoring events. Twenty-five recordings per group is also too few to separate the groups with confidence. Some of the non-confirmatory recordings came from patients with epilepsy, and any label derived from expert review carries that reviewer's misses, so discharges the reviewer missed would push the non-confirmatory group towards the confirmatory one on both statistics. What the external test supports is that the top of the score ranking carries some signal about the recording-level label, and no more than that.

Comparison with other models

Direct comparison against other detectors is limited. Studies differ in recording length, in how close a detection must be to an annotation to count, and in their false positive budget. Our figures come from 21.1 minutes of short epochs.

Scheuer et al. (2017) report the Persyst P13 detector at 43.9% sensitivity and 1.65 false positives per minute on prolonged recordings. Our sensitivity of 0.72 came at 8.5 false positives per minute and fell to 0.28 at ≤ 1 per minute, so the higher sensitivity is bought at the cost of specificity. Deep learning models still perform better. Cyclops reached an internal AUROC of 0.950 against our 0.87, though it fell to 0.846 on external recordings, and SpikeNet was reported at expert level. Both however were trained on far more data: 9,571 recordings for SpikeNet and several thousand patients for Cyclops, against our dataset of 100 recordings.

Limitations

This study has a few limitations. Firstly, the comparison between the amplitude and phase conditions does not cleanly separate shape from timing. The warping functions contain only the stretching or compression needed to align the curve to a template. Other temporal properties, such as duration and asymmetry between the ascending and descending phases, remain in the registered curve. The amplitude and phase conditions are therefore not complementary halves of the waveform, and a null result on the phase component does not establish that timing is uninformative in general. Testing that claim would require a comparison in which the temporal and morphological properties are separated by construction rather than by the registration.

Next, the dataset does not record which channels each marked discharge appeared on. We derived the representative channel from the data but could not check it against the annotation; for the same reason, we could not measure how well consolidation grouped candidates into events. Layer two is therefore unevaluated, and we compute the spatial features in layer four from its output.

Furthermore, the candidate pool reaching the classifier is roughly 98.5% negative. Class weighting balances the contribution of each class to the fit, but it does not increase the number of distinct discharges the classifier sees, and 54 positive windows is a narrow sample of the morphological variation IEDs display.

Finally, in layer 3, each candidate window is divided by a background amplitude estimated once over the whole recording, so that the representation reflects the shape of a transient rather than its absolute voltage. This is stable over a ten-second epoch but not over twenty minutes, when the background shifts with vigilance state and could have affected the detection rates seen on the external testing set. Estimating the background locally, from the period surrounding each candidate, would scale each transient against the background it actually occurs in.

Scope for further study

Each functional principal component is a curve on the same time axis as the waveform, so a component can be plotted and read directly: it shows which part of a transient varies along that

axis, and a score decomposes into how far a candidate moves along each one. A component that separates discharges from mimics therefore describes how they differ, and that description is available here in a way it is not in detectors that return a score alone.

Which components carry discriminatory information remains open. FPCA orders components by the variance they explain, not the amount of discriminatory information they carry, so the ordering says nothing about which of them separate the classes. Methods that order components by discrimination instead, such as functional partial least squares, would build the representation around that separation, and may reach the same performance from fewer components. Those components could then be plotted and read as a description of how IEDs differ from other transients.

Each candidate is scored from the curve on its representative channel, and the spatial features enter only as two scalar quantities in layer four. Multivariate functional data analysis (a method developed and tested by Rigueira et al. (2025)) would instead treat an event as a set of curves, one per electrode, and model them together, so that the way a discharge rises and falls across neighbouring electrodes becomes part of the object being classified. Given that the spatial features raised sensitivity from 0.56 to 0.72 as two scalars, the electric field likely contains more information than those two numbers capture.

With further development, the pipeline has two plausible uses. The first is reviewer assistance, where the detector ranks candidate transients, and experts review the high-scoring candidates rather than reading the whole recording. Lodder et al. (2013) and Jing et al. (2016) built their models for the same purpose, with Jing et al. (2016) reporting a reduction in annotation time of roughly 70%. The current precision supports the same use here, since roughly one in six flagged events was a discharge. The second is autonomous detection, which would require the false positive rate to fall considerably further.

Conclusion

This study applied functional data analysis to detect interictal epileptiform discharges in scalp EEG. By representing candidate transients as curves and separating their shape from their

timing through registration, the functional representation ranked discharges above non-epileptic paroxysmal events better than the scalar baseline (peak-to-peak amplitude). We found that the registered curves carried discriminatory information. Placed within a staged detector, the representation reached a sensitivity of 0.56 at a budget of ten false positives per minute, rising to 0.72 once spatial features were added, with a median localisation error of 12 ms. On a second clinical dataset, the number of detections did not separate recordings in which experts had identified discharges from those in which they had not, although the highest scoring events in each recording did. The representation was fitted from 100 recordings, and each component is a curve on the same time axis as the waveform, so the score can be plotted and read. This distinguishes the approach from detectors that return a score without describing the waveform's morphology.

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Appendix

Declaration of AI use

AI tools were used to assist with adapting Figure 1 from its original source, identifying relevant papers for the literature review, coding assistance, and LaTeX formatting. Claude was used for coding assistance and paper identification. ChatGPT was used for figure adaptation and LaTeX formatting.